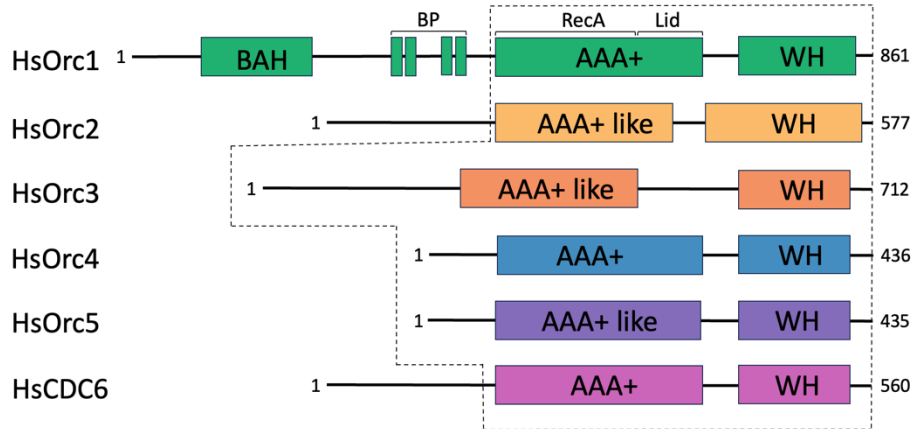


Figure 1 Supplement

A



B

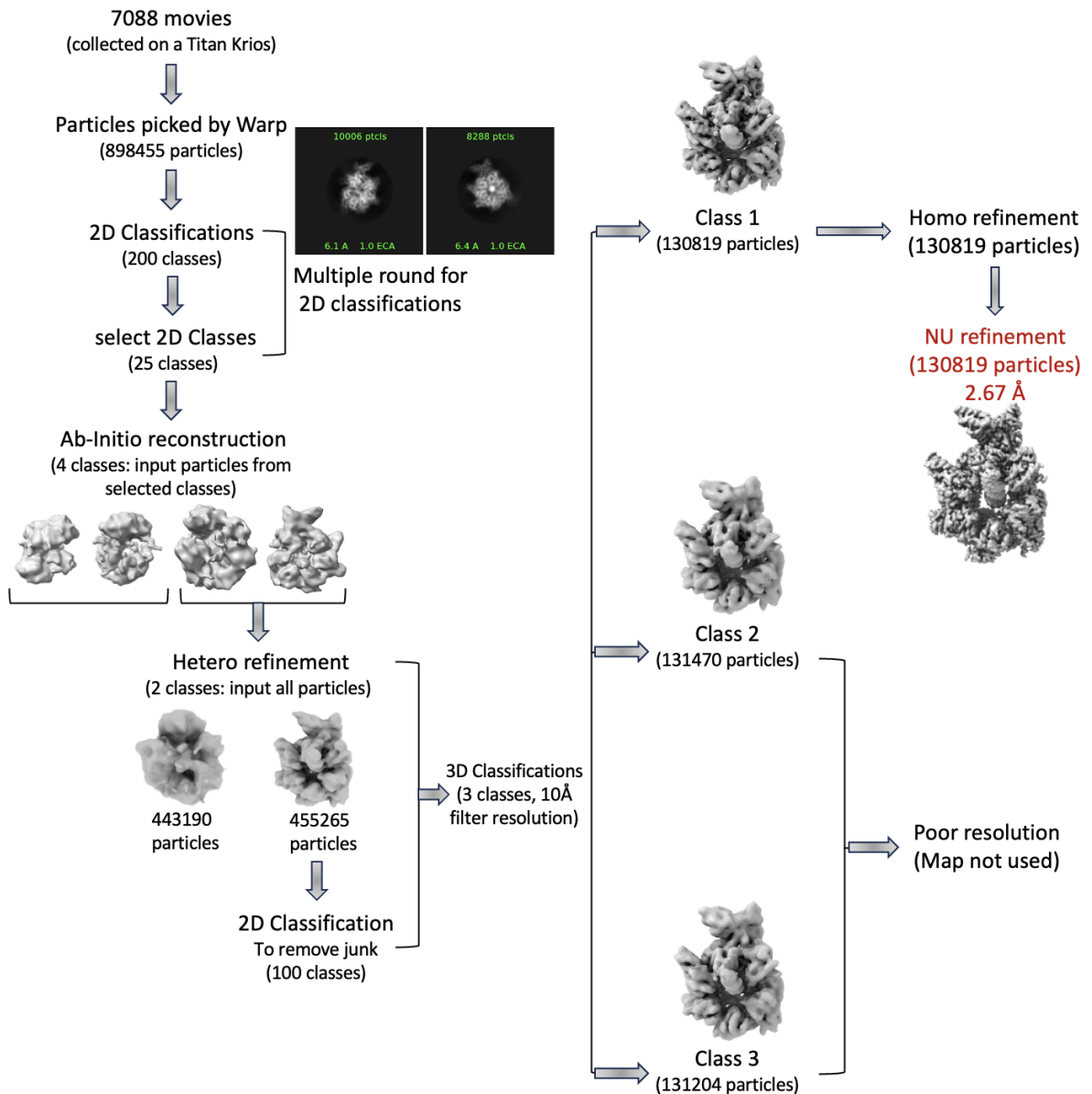
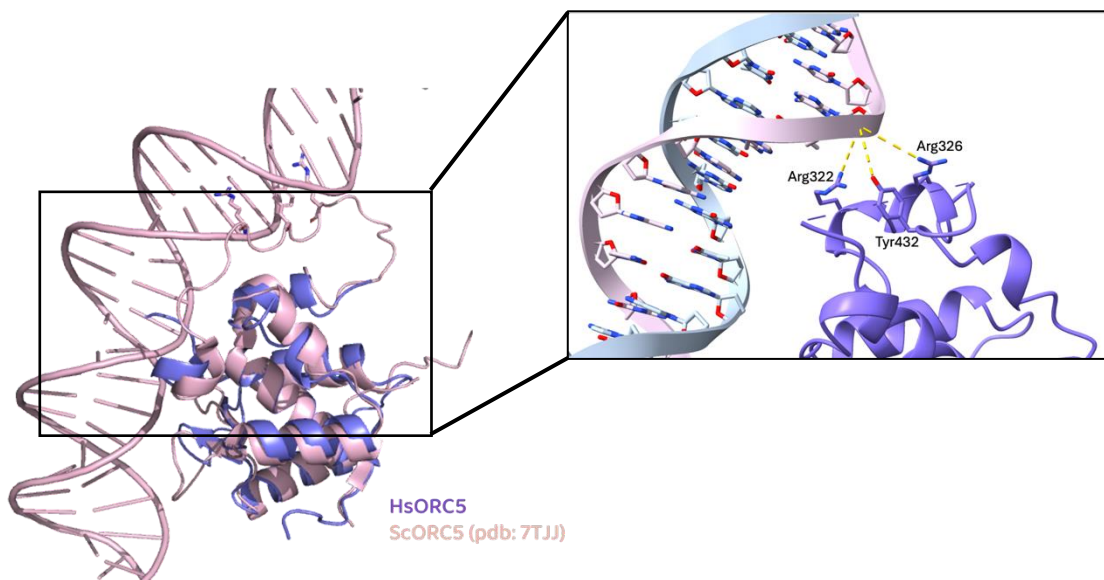


Figure 1 Supplement (Cont'd)

C



D

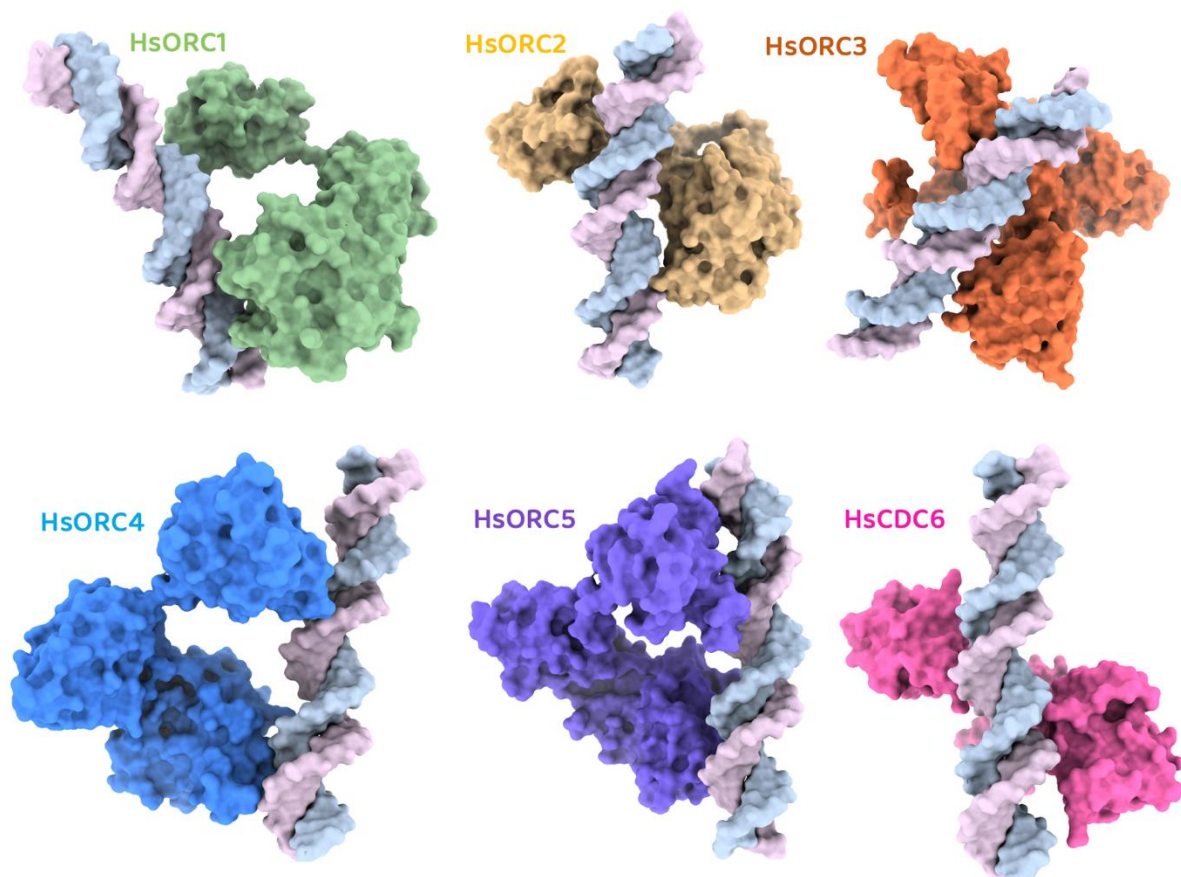
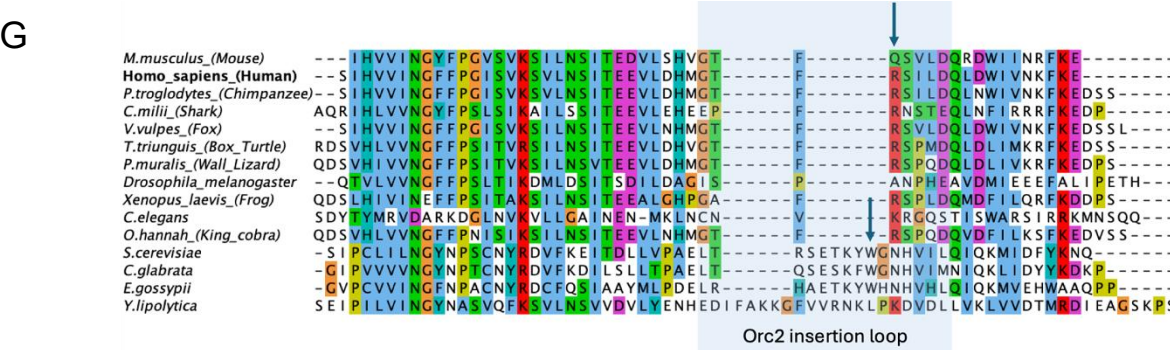
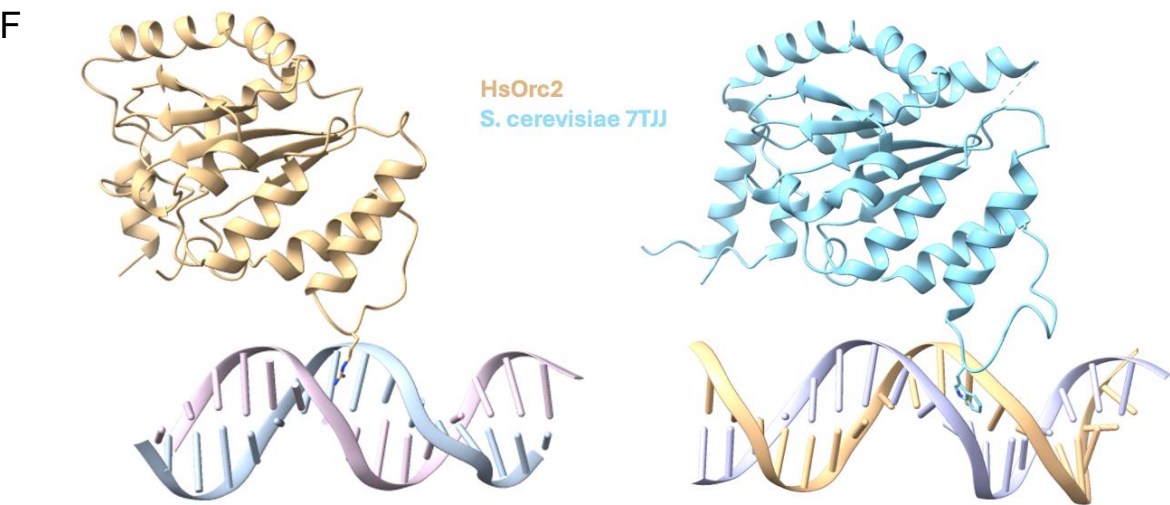
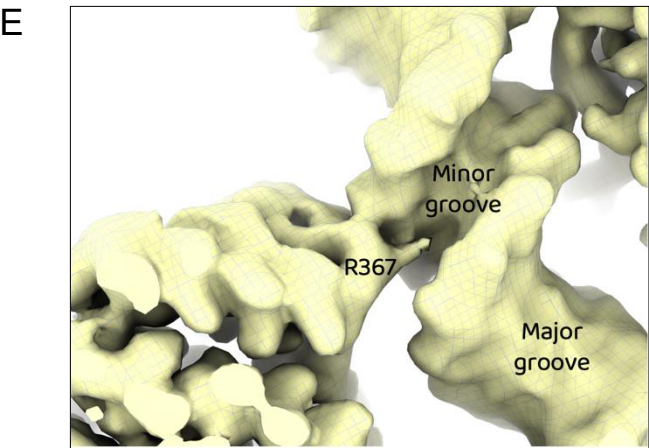


Figure 1 Supplement (Cont'd)



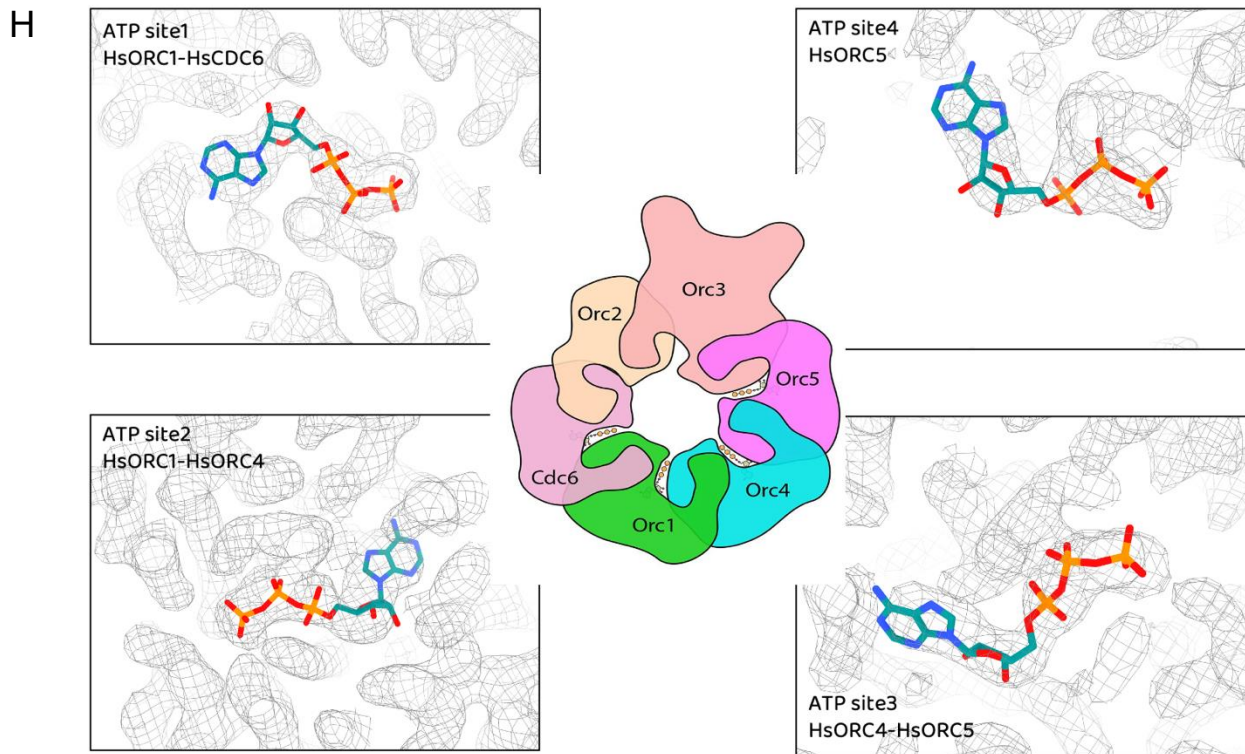


Figure S1 (related to Figure 1). (A) Domain organization and subunit-DNA interactions in the HsORC-Cdc6 complex. Schematic representation of the domain organization of all HsORC subunits (Orc1-5) and HsCdc6. Protein regions that were resolved and modeled in the cryo-EM structure are outlined with dashed gray lines. **(B)** Cryo-EM workflow of HsODC structure. Motion correction, averaging, CTF estimation, and particle picking steps were performed on movies in real time via the WARP (v1.0.9) software package. To determine an *ab initio* reconstruction, 2D classification and all further processing steps were performed in Cryosparc v3.2. See Methods section for details. **(C)** Structure alignment of the human ORC5 subunit with the *S. cerevisiae* DNA-bound ORC5. Alignment shows that *S. cerevisiae* ORC5 contains an extended, flexible loop enriched in basic residues that engages DNA and supports pronounced DNA bending, whereas the corresponding region in human ORC5 is disordered, resulting in DNA contacts limited to the basic patch residues shown in stick representation on the right. **(D)** Structural views of individual protein subunits interacting with DNA, derived from the cryo-EM density map. Figures were generated using ChimeraX. **(E)** Close-up view of the density map around residue R367 from the HsORC2 subunit. R367 is observed making direct contacts with DNA bases within the minor groove. **(F)** Side-by side structures of HsORC2 (yellow, this work) and ScOrc2 (PDBID 7TJJ) (cyan) with R367 from human and W396 of *S. cerevisiae* in stick representation. **(G)** Sequence alignment of the region surrounding R367. Arrows indicate conservation of R367 and the equivalent residue, W396, in *S. cerevisiae* that contacts DNA specifically. **(H)** Cryo-EM density surrounding the ATP molecules at all four conserved ATP-binding sites within the HsORC-Cdc6 complex. The density is well-resolved, allowing precise placement of ATP and modeling of coordinating residues.

Figure 2 Supplement

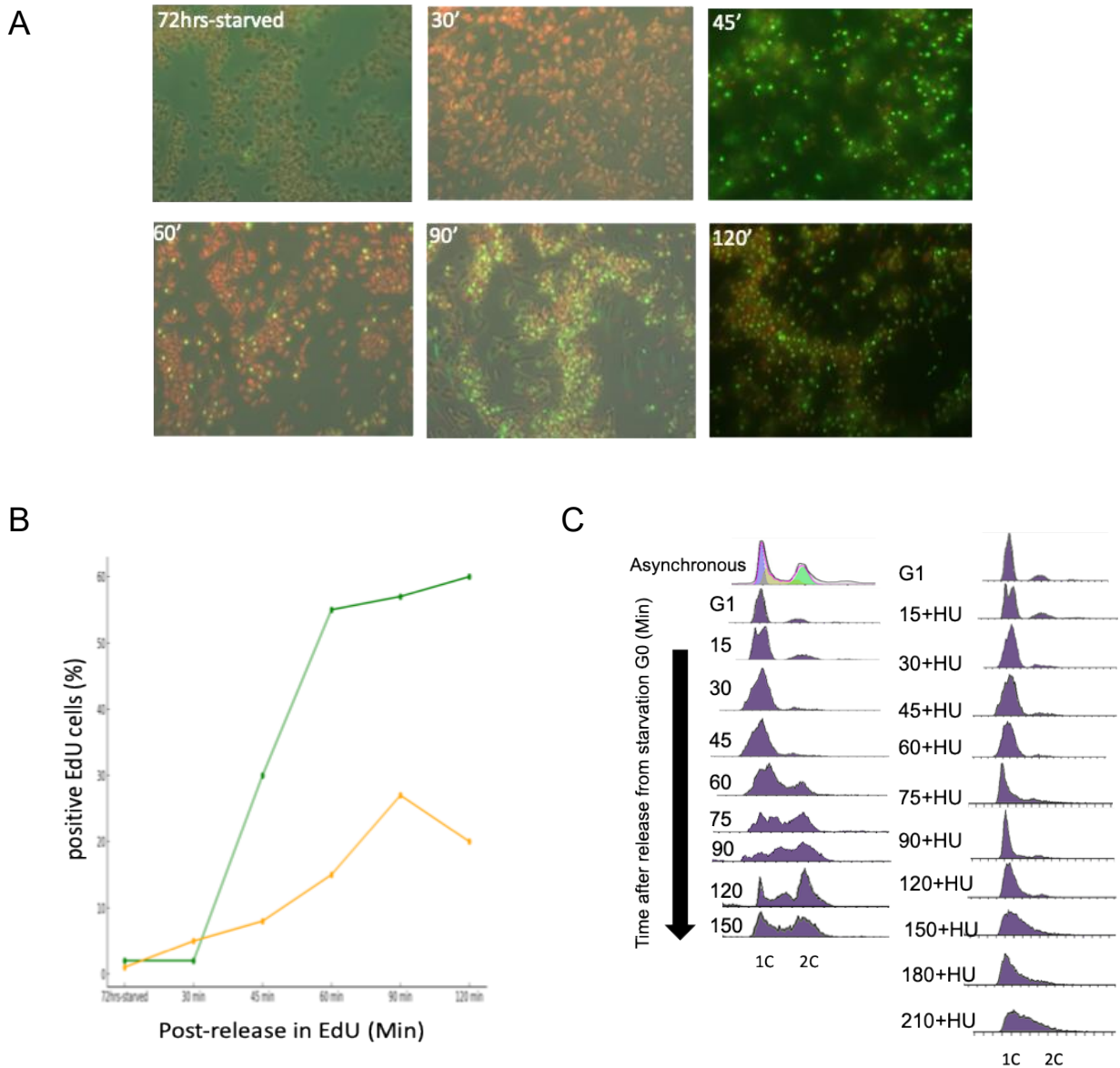


Figure S2 (related to Figure 2). Time-Course Analysis of S-Phase Progression in the Tk+ Strain Using EdU Incorporation. (A) Fluorescence microscopy images cells either starved for released into fresh, complete media for various times were treated with 100 μ M EdU and DNA synthesis was visualized using Click-iTTM chemistry with Alexa FluorTM 488 (green fluorescence). Red fluorescence (DAPI) was used to stain total nuclear DNA for cell visualization. **(B)** The percentage of EdU-positive cells at each time point (Green line) and percentage of budded cells (orange line). **(C)** flow cytometry of DNA content in asynchronously growing cells, cells arrested in G1 by starvation for 72 hours and cells released for different times in the absence (left) or presence (right) of 5 mM HU.

Figure 3 Supplement

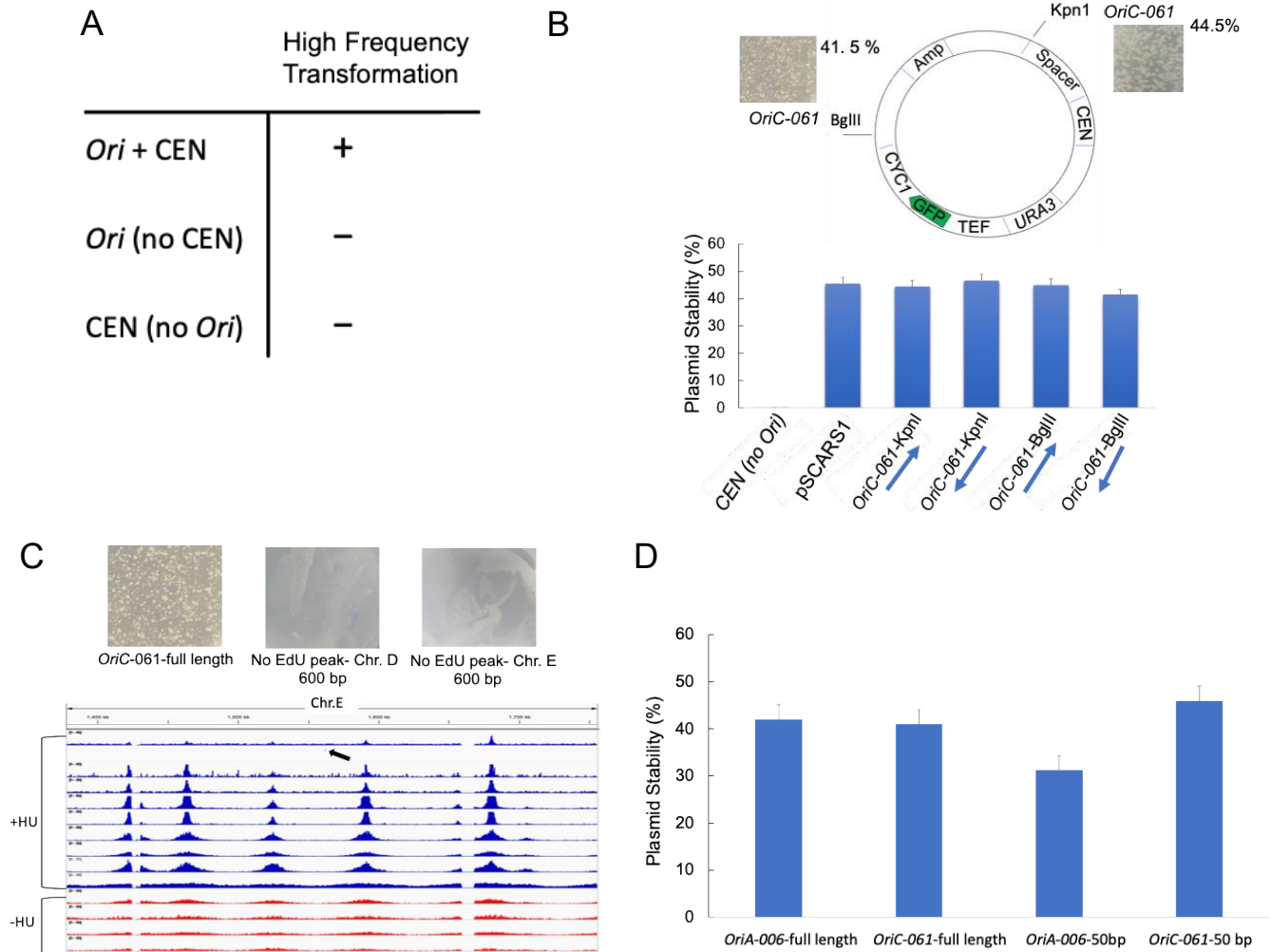
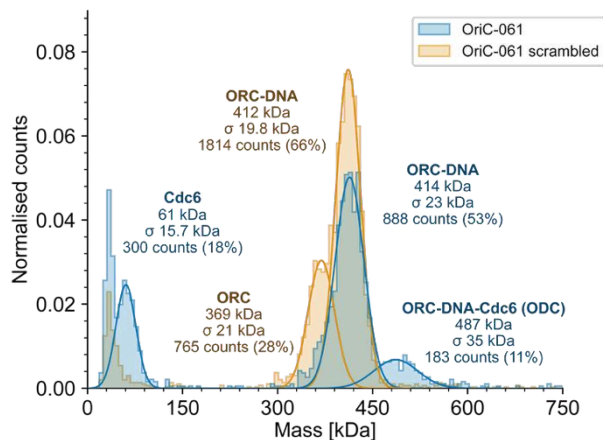


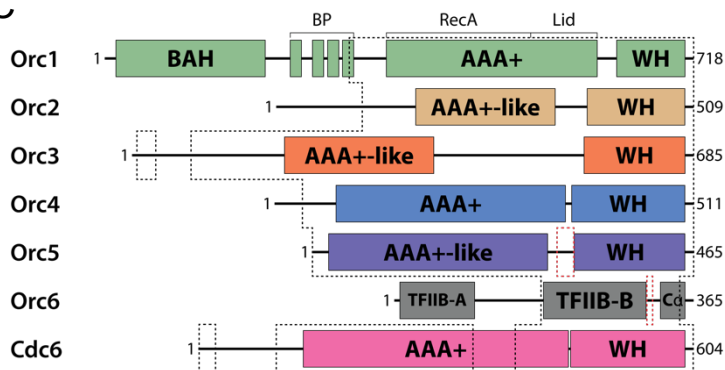
Figure S3 (related to Figure 3). Plasmid assay for origin activity. (A) in *Y. lipolytica*, autonomous replication of a plasmid required both a centromere (*CEN*) and origin of DNA replication (*Ori*) Vernis et al. DOI: 10.1128/mcb.17.4.1995. The plasmid has the *URA3* selectable marker. **(B)** Plasmid construct showing a *CEN*, and two *Ori* insertions sites at either *Kpn1* of *BglIII*. When *OriC-061* was inserted at either site, high frequency transformation occurred and the plasmid stability was comparable. **(C)** The *OriC-061* origin, but not a 600 bp form Chromosome D or Chromosome E (arrow showing region between two origins) show high frequency transformation upon *URA3* selection. The *OriC-061* plasmid is the same as shown in panel B. **(D)** 50 base pair region containing the core 30 base pair region of *OriA-006* or *OriC-061* function as efficiently as a full length 600 base pair region of the genome. The graph shows percentage of plasmid stability and bars represent mean $\pm$ SD from three independent replicates.

Figure 4 Supplement

A



C



B

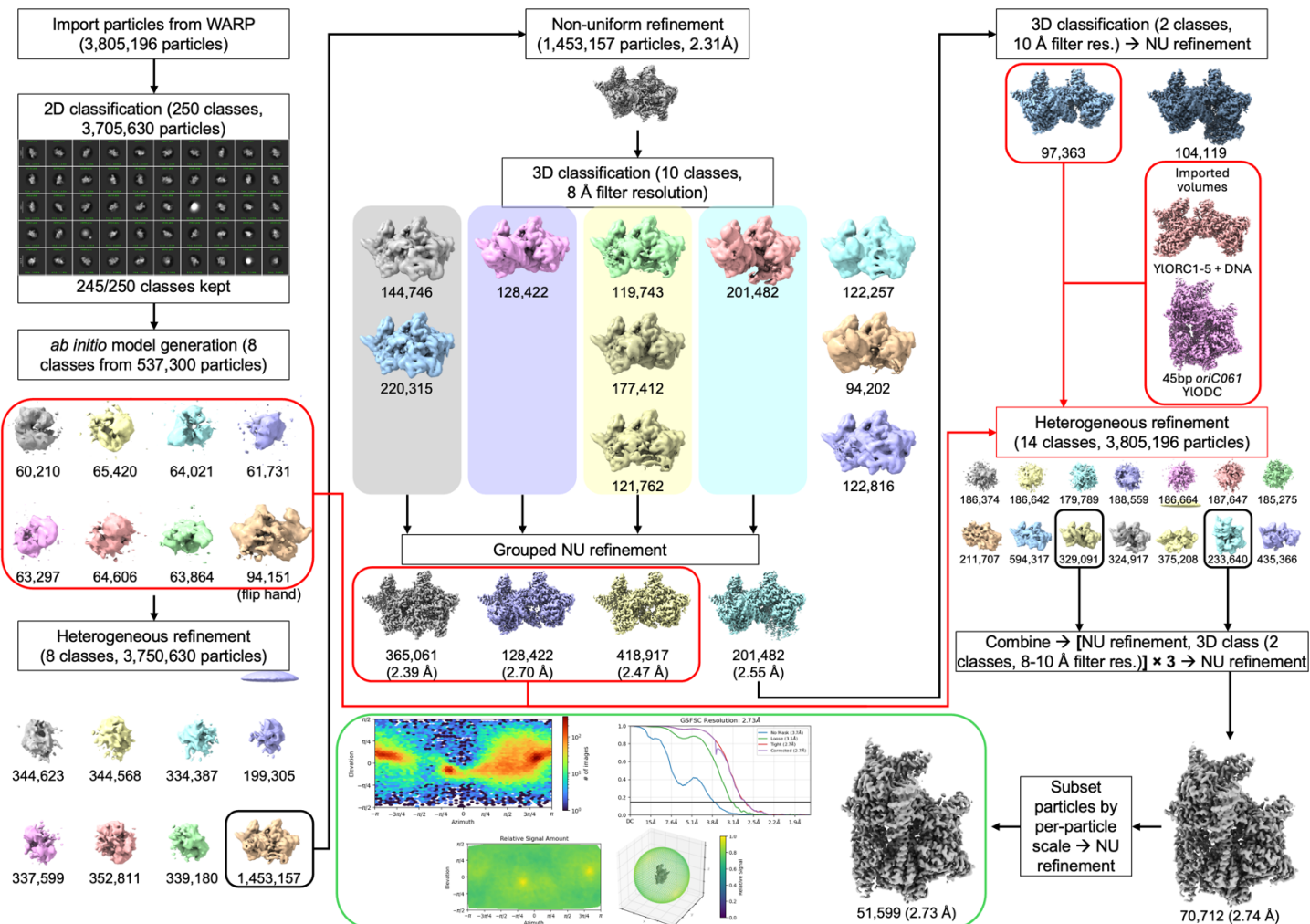
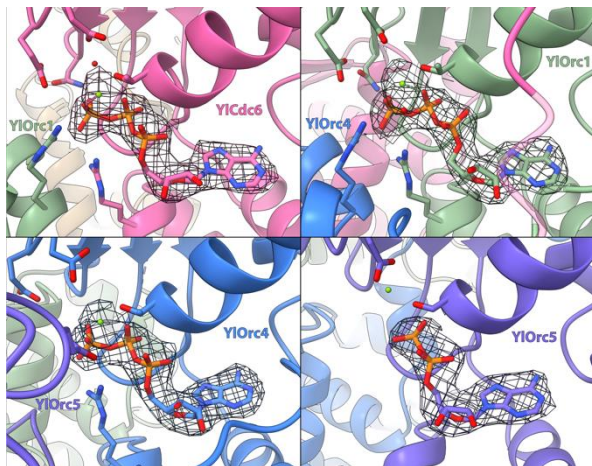
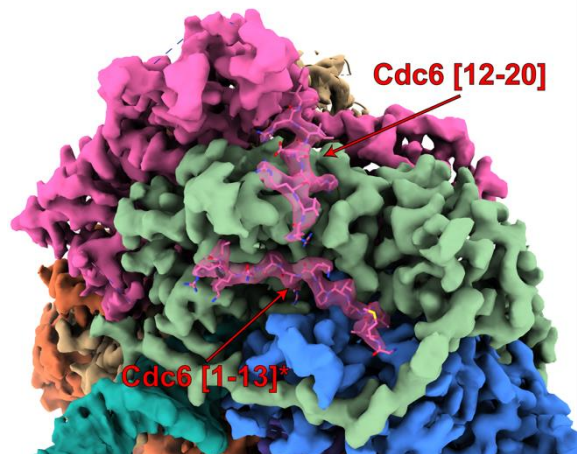


Figure 4 Supplement (Cont'd)

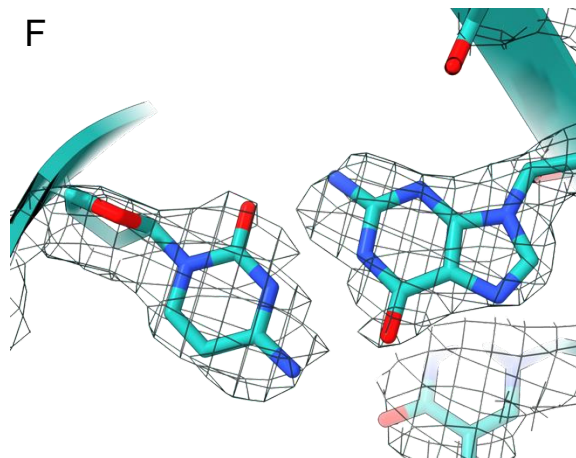
D



E



F



G

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Starmerella_bacillaris_Orc4/1-424
Sugiyamaella_lignohabitan_Orc4/1-161
Blastobotrys_adeninivorans_Orc4/1-443
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Komagataella_pastoris_Orc4/1-578
Candida_albicans_Orc4/1-564
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Torulaspora_delbrueckii_Orc4/1-571
Saccharomyces_cerevisiae_Orc4/1-529
Tortispora_caseinolytica_Orc4/1-370
Tuber_brumale_Orc4/1-413
Didymosphaeria_variable_Orc4/1-744
Neurospora_crassa_Orc4/1-1021
Schizosaccharomyces_pombe_Orc4/1-972
Drosophila_melanogaster_Orc4/1-459
Danio_reio_Orc4/1-432
Xenopus_laewis_Orc4/1-432
Homo_sapiens_Orc4/1-436
Arabidopsis_thaliana_Orc4/1-417

Orc4 insertion loop	
437 RIWSRGVARAAWEKLESNLLSPVEKSAKRVAQLA-----SDTSLDDEIRMTTRVDVSLQELTNML497	
495 RVVSRDVARSAWERLENLDLHVETPGSNTQGSITATATGLVGDSKATGTVRRDLKMAKVDVNLLEIGDIIS566	
363 TKISILLAKHAWERLINLGLLPV-----NANDGFSLEKMYVPELNIDDLKLL411	
91 RIWSRGVARSAWEKLEAMLVVYISSTATGKAAA-----AGAALDDELKLVVDASLLEISNMI149	
376 RIWSKEAARAWEKLEAMDLILTINSAPSKA-----PTSAVADIKMVVDAGLLELGHMI431	
376 RIWSKDVARSAWERLEEFDLVPLSTNTK-----GTDAARRDIRMVVDGLLELGNMI429	
505 NVYKFLMEKCWETLITKYLLLDINVSSRNTNKSQSFMF-----NNNLSRSRVKFFYLVNVTLEELSEIL568	
488 KVLSSRVSLNWSSTLYKLGILLDPKGDYVEGQSNNSYHI-----NNKLLIEENDMVLDITLDELNGFI552	
468 MWTKHDMKNVWENLANOELITERGAISVRFTEDSAIQSANY-----NSSTMKAPFDLTYVVCVTLOEMRLI536	
491 KLWKKDVKVNVWETLINLNFLEKSAVGLRSEAIAVFYASNY-----QLQGTTFIPFDLALYQTHVTLSDLRRIIV559	
449 KWLWKDVKVNVWENLVQLDFEFTEKSAVGLRDNATAAFYASNY-----QFQGTMIIPFDLSYQMIILLOELRRII517	
338 HINARDPARNK-----KICMGKLTSTCA359	
341 RLWSKEVALGAWERLAELFLLVPAVIGGG-----SGGGVSRFCRMWRVEVGLAEIGKCL394	
679 VVWCKEVARREWEGLLELELYMPA-----VGGGLTGFSMVKCDVSLLEEVGAVL727	
942 VVWGRGAGIAWERLVTGLLVPAAGSLGGGAGANARGAGATL-----GQSGGGLGCMWRVDVALEEPGAV1008	
901 RLGRDVALEAWETLGSGLIIPVHPSTN-----VSGALSRCQLWQPEVDINVSVDL954	
388 --VERSIVLKAFEHLRIAEIMPLTGGAGG-----GVGKVVQKEFMHKLALTYSDIHHCM440	
365 --FEKPVVMKAFEHLLQLLVRPVDS-----GVCKVQRVQLMRLMLEHGMVMEAL413	
365 --FEKAVVIAKAFEHLLQLLVRPVDS-----LSVRTQKEVRLMKLLDNTDIVEAL413	
367 --FEKPVVMKAFEHLLQLLVRPVDS-----TSNNSQREYQLMKLLDNTDITMNAL415	
353 --ALNVCLRAFEHLRERQVICAEN-----RGQSQTGEYRLQKLLLSASELHQGM401	

H

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Geotrichum_candidum_Cdc6/1-671
Starmerella_bacillaris_Cdc6/1-511
Sugiyamaella_lignohabitan_Cdc6/1-633
Blastobotrys_adeninivorans_Cdc6/1-537
Trichomonascus_ciferrii_Cdc6/1-556
Komagataella_pastoris_Cdc6/1-461
Candida_albicans_Cdc6/1-481
Kluyveromyces_lactis_Cdc6/1-523
Torulaspora_delbrueckii_Cdc6/1-500
Saccharomyces_cerevisiae_Cdc6/1-513
Tortispora_caseinolytica_Cdc6/1-555
Tuber_brumale_Cdc6/1-612
Didymosphaeria_variable_Cdc6/1-637
Neurospora_crassa_Cdc6/1-691
Schizosaccharomyces_pombe_Cdc6/1-577
Drosophila_melanogaster_Cdc6/1-662
Danio_reio_Cdc6/1-561
Xenopus_laewis_Cdc6/1-554
Homo_sapiens_Cdc6/1-560
Arabidopsis_thaliana_Cdc6a/1-539
Arabidopsis_thaliana_Cdc6b/1-505

Cdc6 specificity loop	
529 VCSALEYSVINITGICGKNL-GI----N-GKGRASGGTGASKGEV--YGIRDDYVQRKVTLVNQRMDDI591	
590 VIAALEANGVNVNVCVCRKGLRAK-----DSDRVRTSGGTGAAILVRT--LSGREDSKRRIAANVHMDL654	
433 VVNALAAAGLVAVTCLCFKKGL-GV----H-GSNRISRGGSASSAKGA-DGCRDDFGGLRKLSSRAQLLDV496	
552 VISALEFGSGVITVSGVCRKGL-GV----T-GRRRTSRGGGASAGQST-LGSRREYQRRKLSSNVHRRVDL615	
456 LISALEFGSLVTISGVCRKGL-GV----AQSRGRVSRGGSCATSSKSSYLGAARDYSQRRKLSSNVHRLDM521	
480 VISALEAGVNVNVCVCRKGM-----GSARGRSSGGTGAGAAK--PAARDYSQRRKLSSNVHRRMDL540	
409 LLSMLEHGVISM-----GNSSSLSNVG-----NSKAVACITAEHEI445	
428 LLYALESSCILL-----TNSNKNNGN-----SNKIKLVNVPYDDI463	
472 SCYVLSGSGVITVSGVCRKGL-GV----EFTKRLLPVSG-----EFTKRLLPVSG-----EFTKRLLPVSG509	
447 LCNALREGGVITVSGVCRKGL-GV----GRSRGRKTRH-----GRSRGRKTRH-----GRSRGRKTRH484	
461 ICTILECGLVSIKK-----TKCKGKTKM-----TKCKGKTKM-----TKCKGKTKM498	
507 MNALESAGIV-----NRILSRHGT-----NRILSRHGT-----NRILSRHGT540	
549 VVNSLELVAGVNVNVCVCRKGM-----GGMGTTPSKKGRASGGM-----GGMGTTPSKKGRASGGM593	
562 IVSGLETLSSLVSAVE--GRAGSLVA--VNTPSRKSCKGFGASV--VNTPSRKSCKGFGASV--VNTPSRKSCKGFGASV615	
598 VVSGLETLSSLITIPVD--GKTGSFTALGYSSPGVGTGPKRGGKKKKDVGFM--GGALVAG--GGALVAG--GGALVAG665	
525 VANSLETLAIIIRL-----RTKQRNGKP-----RTKQRNGKP-----RTKQRNGKP540	
608 TVDLVETRGILRI-----MRKKEPK-----MRKKEPK-----MRKKEPK560	
510 LCSLLELRGILTL-----KKAADAR-----KKAADAR-----KKAADAR542	
504 LCQLLELRGILGL-----KKAADAR-----KKAADAR-----KKAADAR536	
509 LSGLLARGLGL-----KKNKETK-----KKNKETK-----KKNKETK541	
492 MCTVLDNQDILKL-----SLRADDK-----SLRADDK-----SLRADDK524	
458 MCTVLDNQDILKV-----GLARDK-----GLARDK-----GLARDK490	

Figure 4 Supplement (Cont'd)

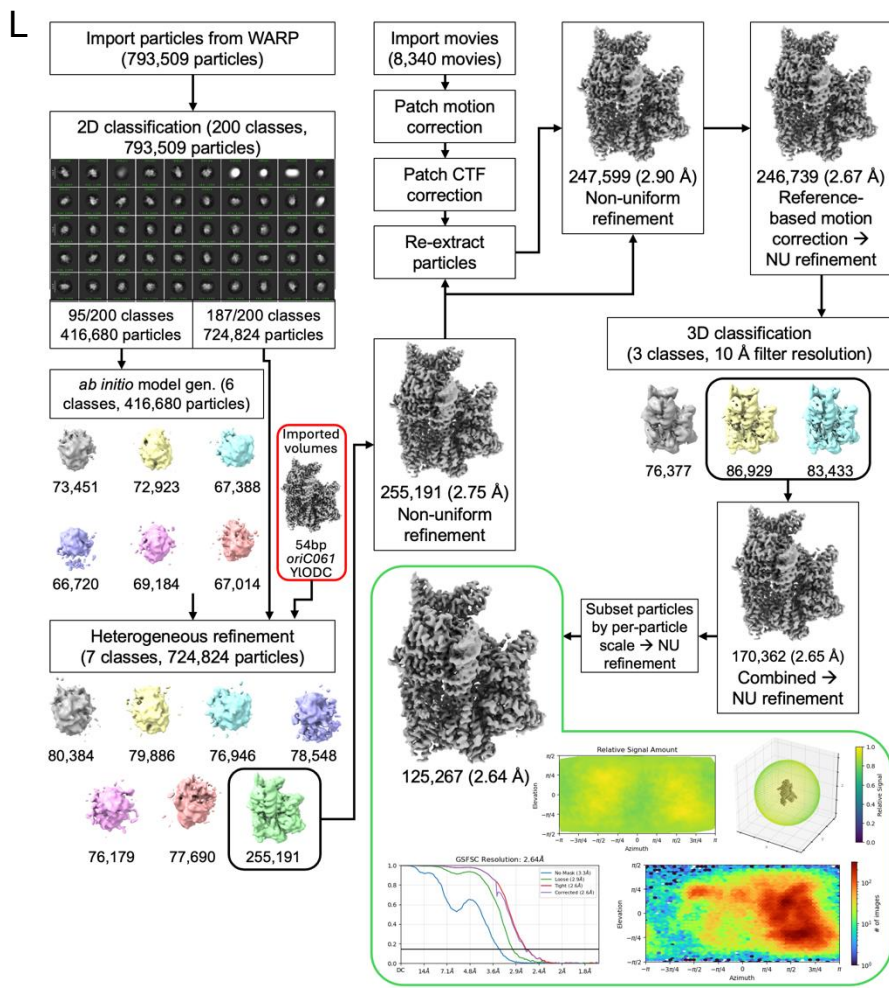
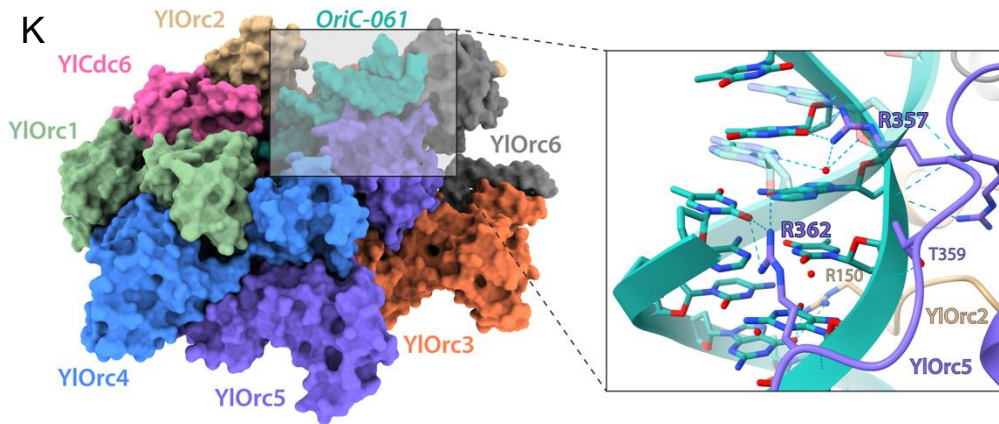
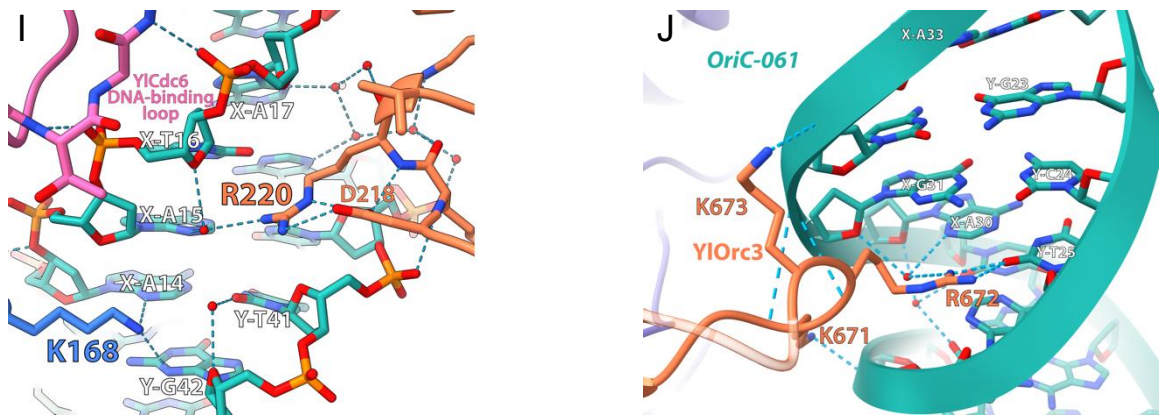
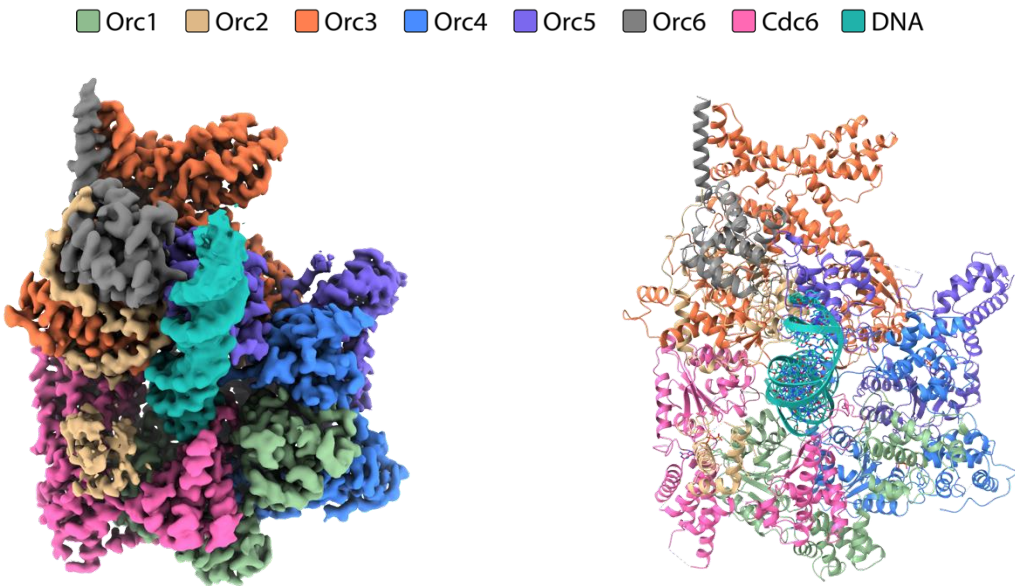
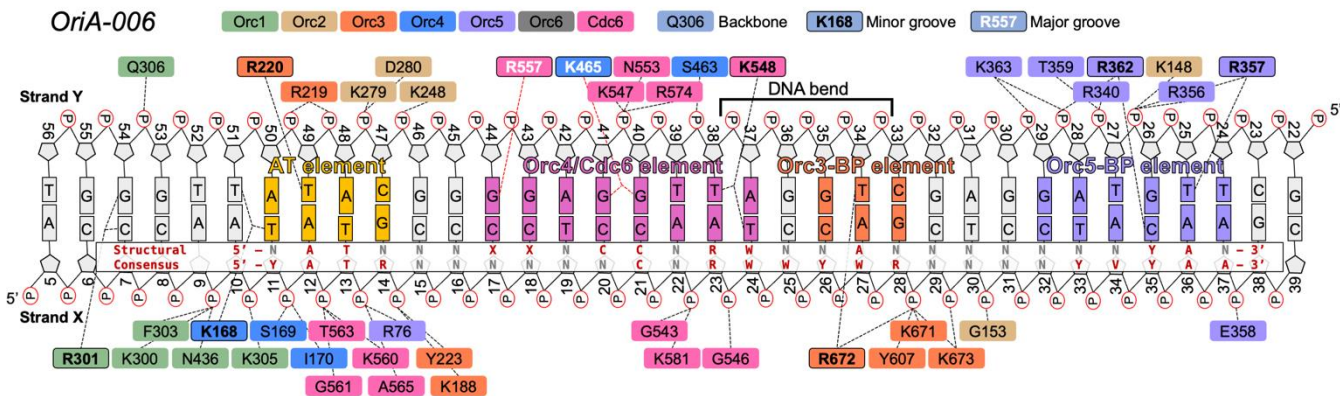


Figure 4 Supplement (Cont'd)

M



N



O

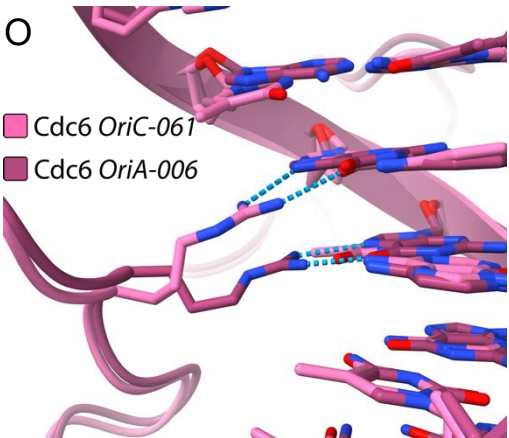


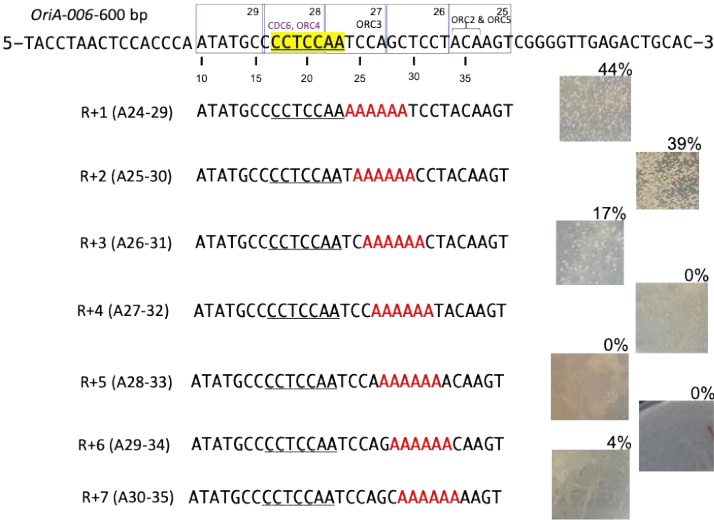
Figure S4 (related to Figure 4). **(A)** Mass photometry data of the gel filtration peak fractions from the loading assay in Figure 4A. **(B)** Schematic diagram of the cryo-EM data processing workflow and statistics from the generation of the YIORC-DNA^{54bpOriC-061}-YICdc6 cryo-EM map. **(C)** Schematic of the domain organization of YIORC and YICdc6. Dotted lines surround the regions visible within either YIORC-DNA-YICdc6 maps. **(D)** ADP/ATP in stick representation at the four ATP-binding sites within the YIORC-DNA^{54bpOriC-061}-YICdc6 complex, with a mesh overlay of the representative cryo-EM density of these nucleotides. **(E)** A view of the YICdc6 IDR sections binding to the YIOrc1•YIOrc4 AAA domain interface and the YIOrc1-AAA•YICdc6-WHD interface. * denotes the visibility of a leftover glycine on the N-terminus of YICdc6 resulting from TEV protease cleavage of an N-terminal 8xHis-SUMO tag. **(F)** Representative cryo-EM density of a base pair of *OriC-061* in the YIORC-DNA^{54bpOriC-061}-YICdc6 complex. **(G)** An MSA of the Orc4 region surrounding the Orc4 insertion loop including various related yeasts and eukaryotes. Pink boxes in the YIOrc4 and ScOrc4 sequences denote the region that encompasses the species' insertion helix. **(H)** An MSA of the Cdc6-WHD surrounding the specificity loop. The red box indicates species containing a conserved specificity loop, while blue indicates those that appear to lack sequence homology in this region. **(I)** View of the AT element (positions X-A14 to X-A17) of *OriC-061* in the YIORC-DNA^{54bpOriC-061}-YICdc6 structure, displaying the hydrogen bond interactions between protein, DNA, and water. **(J)** Same as **(I)** but of the Orc3 basic patch (Orc3-BP) element of *OriC-061*. **(K)** Same as **(I)** but of the Orc5 basic patch (Orc5-BP) element (positions X-G35 to X-A40) of *OriC-061*. **(L)** Schematic diagram of the cryo-EM data processing workflow and statistics from the generation of the YIORC-DNA^{60bpOriA-006}-YICdc6 cryo-EM map. **(M, N)** Same as Figure 4B and 4C, respectively, but of the 2.6 Å resolution cryo-EM map YIORC-DNA^{60bpOriA-006}-YICdc6 and corresponding structure. **(O)** A view of the difference in the major groove base interaction of YICdc6 R557 between YIORC-DNA^{54bpOriC-061}-YICdc6 and YIORC-DNA^{60bpOriA-006}-YICdc6.

Figure 5 Supplement Figures

A

Origin	Mutant	Orc4-Cdc6 Motif <i>OriC</i> CCNCCNR/ <i>OriA</i> CNNCCNR	ARS activity	Plasmid Stability (%)	Cdc6 Loading Efficiency (%)
<i>OriC</i>	WT	CACCCA	+++	47	100
<i>OriC</i>	C1A	ANCCNA	+++	48.6	NT
<i>OriC</i>	C2A	CNACNA	+++	47.5	NT
<i>OriC</i>	C3A	CNCANA	+++	48.8	NT
<i>OriC</i>	C1,C2,C3	GNGGNA	—	0	NT
<i>OriC</i>	C1,C2,C3	TNTTNA	—	0	NT
<i>OriC</i>	R to C	CACCCC	++	4	NT
<i>OriA</i>	WT	CCTCCAA	+++	47	NT
<i>OriA</i>	C1A	ANNCNA	+++	48	NT
<i>OriA</i>	C2A	CNNCANA	+++	47.3	NT
<i>OriA</i>	C1G	GNNCCNA	+++	47	105.1 ± 1.3
<i>OriA</i>	C2C3G	CNNGGNA	+++	22	88.0 ± 2.0
<i>OriA</i>	C1C2C3	GNNGGNA	—	0	46.9 ± 0.7
<i>OriA</i>	C1C2C3	ANNAANA	—	0	20.3 ± 0.8
<i>OriA</i>	C1C2C3	TNNTTNA	—	0	NT
<i>OriA</i>	R to C	CCTCCAC	++	4	NT

B



C

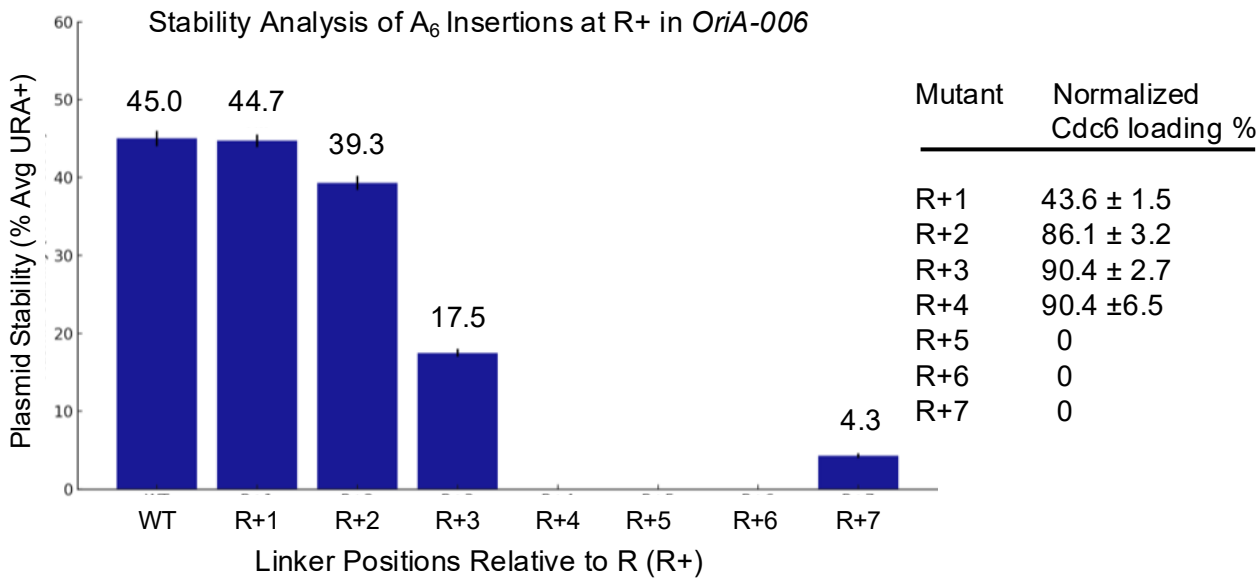
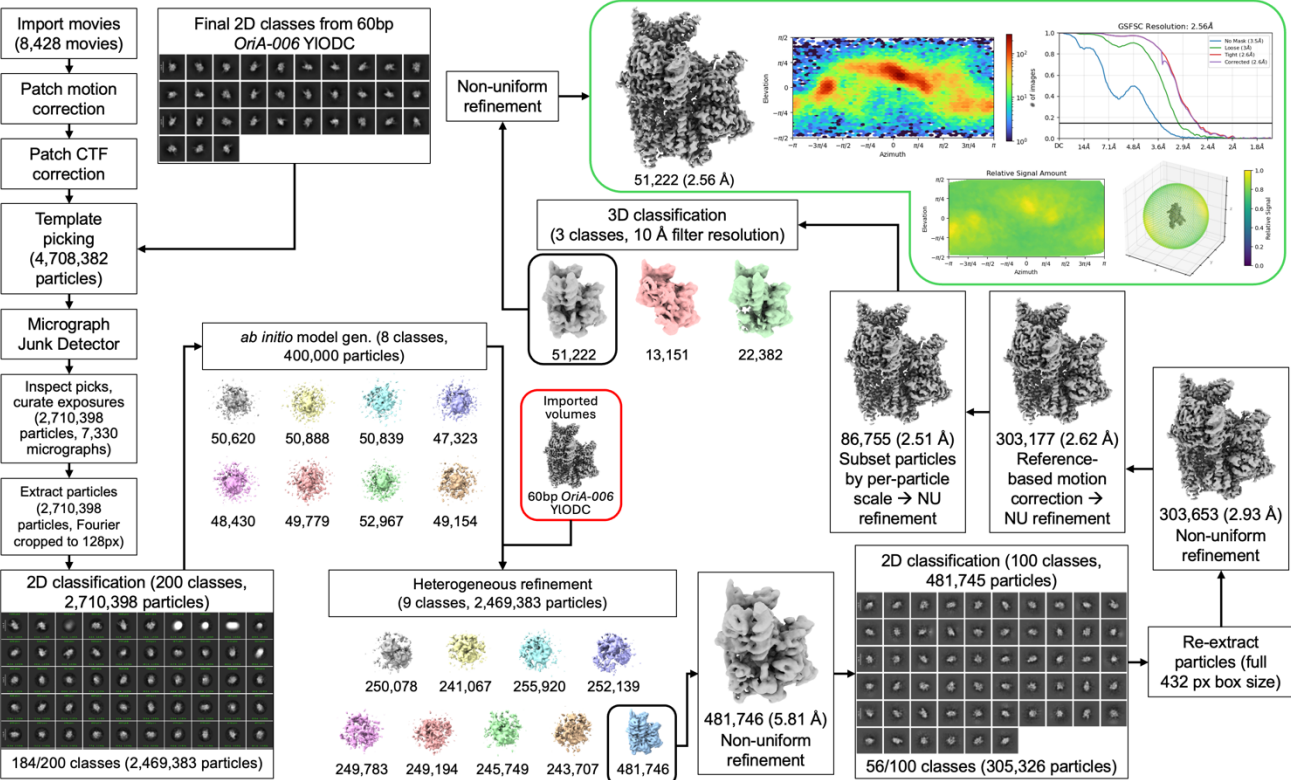


Figure 5 Supplement (Cont'd)

D



E

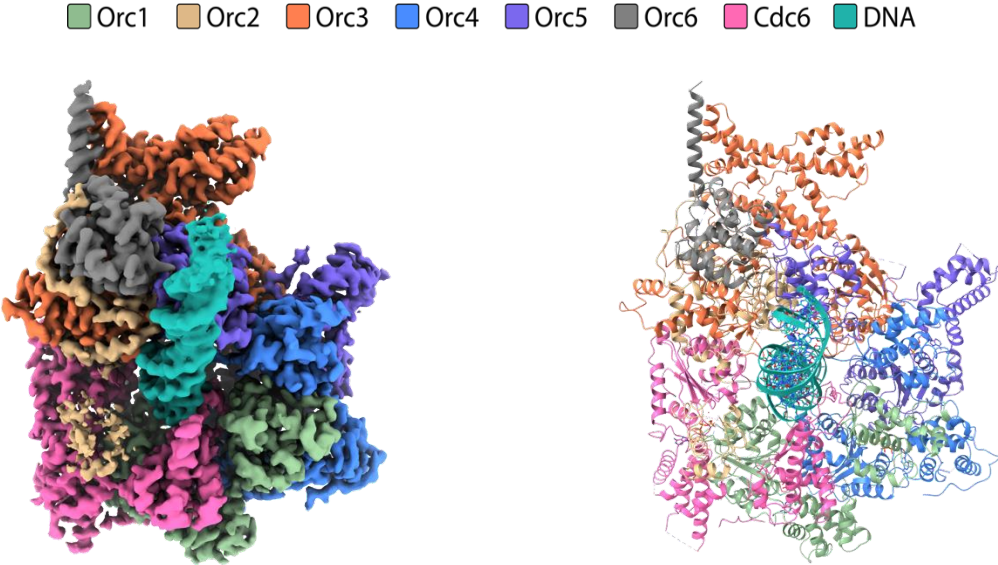
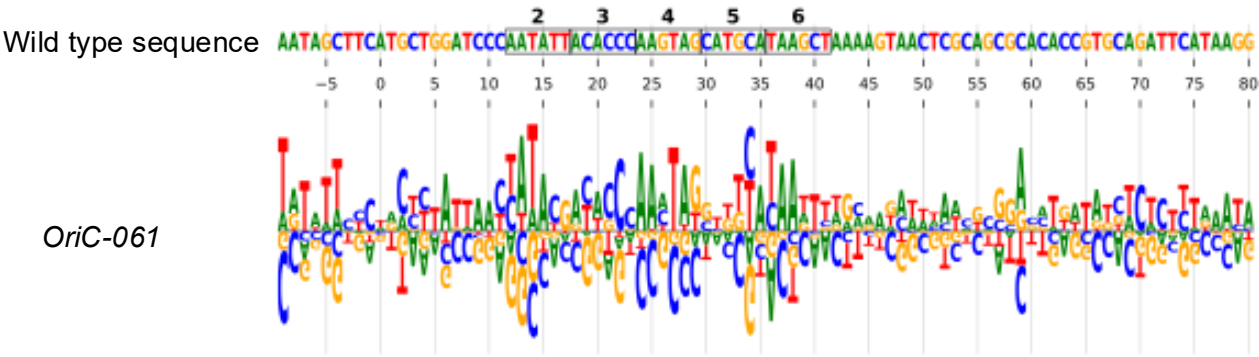


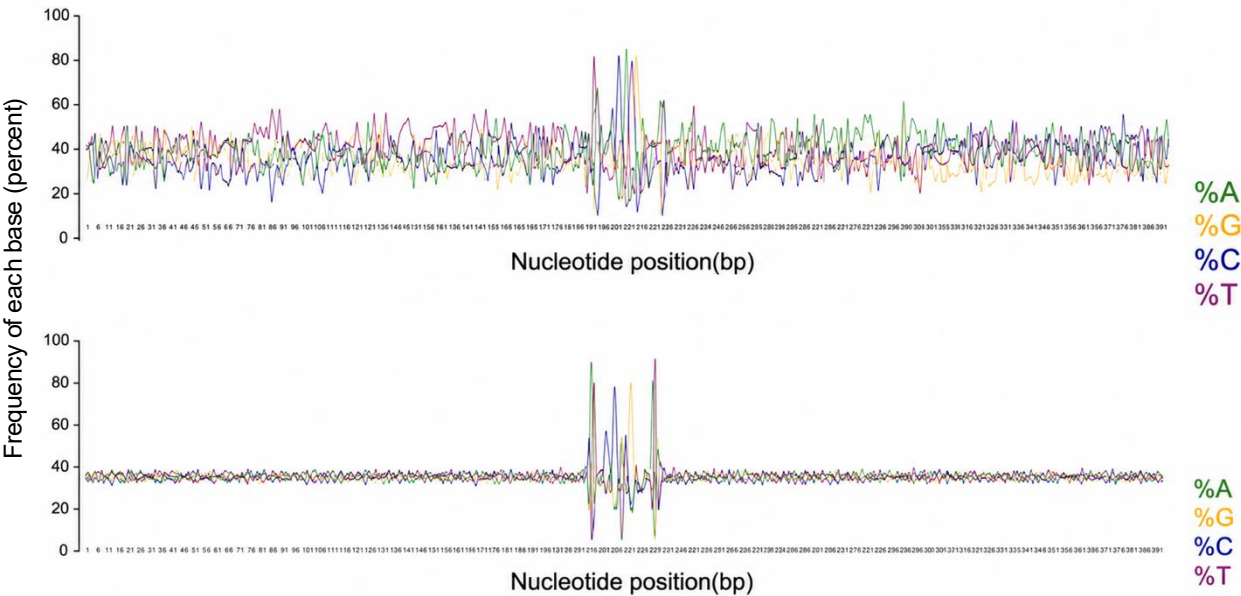
Figure S5 (related to Figure 5). Analysis of *Y. lipolytica* mutant origins. **(A)** Summary table of mutations in *OriA-006* and *OriC-061* reporting high frequency transformation (*ARS* activity) of point mutations and A₆ insertions. **(B)** Analysis of mutant *OriA-006* with insertion of six A residues at different positions. The high frequency transformation images and percent plasmid stability was done as described in Figure 5 legend. **(C)** Summary of the plasmid stability of the A₆ insertion mutations. **(D)** Schematic diagram of the cryo-EM data processing workflow and statistics from the generation of the YIORC-DNA^{60bpOriA-CNNGGNNR}-YICdc6 cryo-EM map. **(E)** Same as Figure 4B and Figure S4L, but of the 2.6 Å resolution YIORC-DNA^{60bpOriA-CNNGGNNR}-YICdc6 cryo-EM map and the accompanying structure.

Figure 6 Supplement

A



B



C

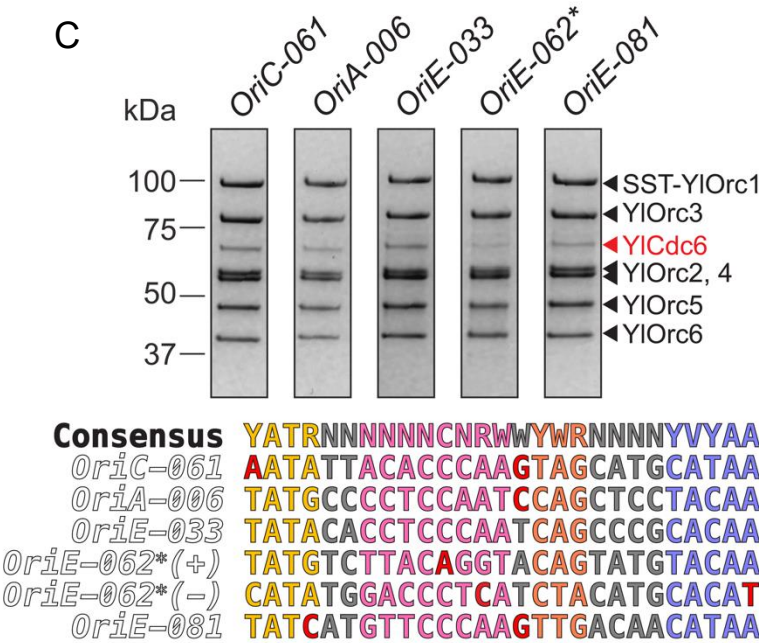


Figure 6 Supplement

Figure S6 (related to Figure 6). (A) Additive model for origin specificity derived from an MPOS assay on 90 bp sequences containing *OriC-061*. Boxed region indicates the core region identified by linker scanning (linker positions 2-6 in **Fig. 3A**). Note that the base numbers refer to the position within the 90 bp and the numbers refer to the numbering of nucleotides in Figure 4C. (B). Analysis of nucleotide frequencies across 54 *Y. lipolytica* early origins of replication compared to 5351 non-origin sequences matched the consensus sequence. The early origins showed a skew for T/A sequences 5' to the core consensus sequence and a skew for A/T base pairs 3' to the core consensus sequence. There is no skew surrounding the non-origin consensus sequences. There are 5545 matches to the consensus sequence in the genome. (C) Cdc6 loading assays carried out on 60bp dsDNA segments containing *bona fide* origin sequences *OriC-061* ($p = 2.12\text{e-}06$) and *OriA-006* ($p = 3.79\text{e-}08$) or presumed origin sequences *OriE-033* ($p = 3.76\text{e-}12$), *OriE-062** (contains two matches on opposing strands, + strand $p = 1.34\text{e-}05$, - strand $p = 1.87\text{e-}05$), and *OriE-081* ($p = 1.19\text{e-}04$), in addition to each origin's comparison to the consensus motif, with sequence elements using the previous coloring scheme and mismatches colored red.